

Keyanna Takita
CS414
Assignment05
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Question 3: Scope

A)

```
#include <iostream>
void f(int y) {                // y is now 2?
    int x = y + 1;             // x=3; 2+1
    for (int i = 0; i < 3; i++ { // loop
        int temp = i*2 + x;
        std::cout << i << " " temp << " " << x << std::endl;
    }                          i = 0, 1, 2
}

int main(int argc, char *argv[]) { // one local variable
    int x = 2;                     // parameter is y
    f(x);                          // change
    std::cout << x << std::endl;
    return 0;
}
```

main x = 2, is nested f x = 3, then is nested i = 0, 1, 2 and temp = 3, 5, 7

Output 2

B)

```
#include <iostream>
int main() {
    int a = 5;                  // what is being reference
    int & ref_To_An_A = a;      // refernce to og a
    {
        int a = 10;            // new / nested
        std::cout << ref << std::endl;
    }
    return 0;
}
```

main, a = 5, then reference to an A, which goes back to a = 5. The nested inner section a = 10 which taken over a = 5 but the output would be 5 still.

C)

```
#include <iostream>
int main() {
    int* p = new int(42);    // int is containing 42 stored in
p
    {
        int* q = p;        // copy addr. From p to q
        std::cout << *q << std::endl;    // same pointer = 42
    }
    delete p;                // p has left the villa
}
```

main: outer section is p which points to 42. The inner section is q, which also points to 42.

Going back out q = 42 while p still points to 42 as well. p is deleted and opens memory.